# Lab 1: Advanced Medical Image Pre-processing Methods

Medical Image Analysis

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# AI Disclaimer and Workload Distribution

| Sections | Completed by |
|---|---|
| Data loading and Characterization | Álvaro Gil Burgo and César M. Robalo |
| Noise addition | Álvaro Gil Burgo |
| Standard Smoothing Methods | Álvaro Gil Burgo |
| NLM Filtering | Álvaro Gil Burgo |
| Anisotropic Diffusion | César Manuel Robalo Fariñas |
| Advanced Filtering Method | César Manuel Robalo Fariñas |
| Comparative Analysis and Metrics | César Manuel Robalo Fariñas |

Supplemental assistance was provided by Generative AI tools (Claude and Google Antigravity) under course rules to help develop this laboratory project. AI tools were specifically used to:

- Helping to debug, optimize, and understand particular code routines and algorithms.
- Clarification of theoretical concepts related to sophisticated filtering methods.

The final code implementation, inline documentation / comments, theoretical interpretations and the complete writing and synthesis of this report was all authored, verified and executed by the group members without automated generation.

## 2. Noise addition

In this practice two noise models were applied to both images. These noise models were the Gaussian model and the Impulsive (salt and pepper model).

**Gaussian noise:** This noise slightly alters the value of every pixel in the image. For each pixel, a random number is drawn from a normal distribution with mean 0 and standard deviation σ. Because the noise follows a normal distribution, most pixels change only a little and a few change a lot, so the image looks grainy. As intensities are normalised between 0 and 1, values pushed outside that range are simply cut off at 0 or 1.

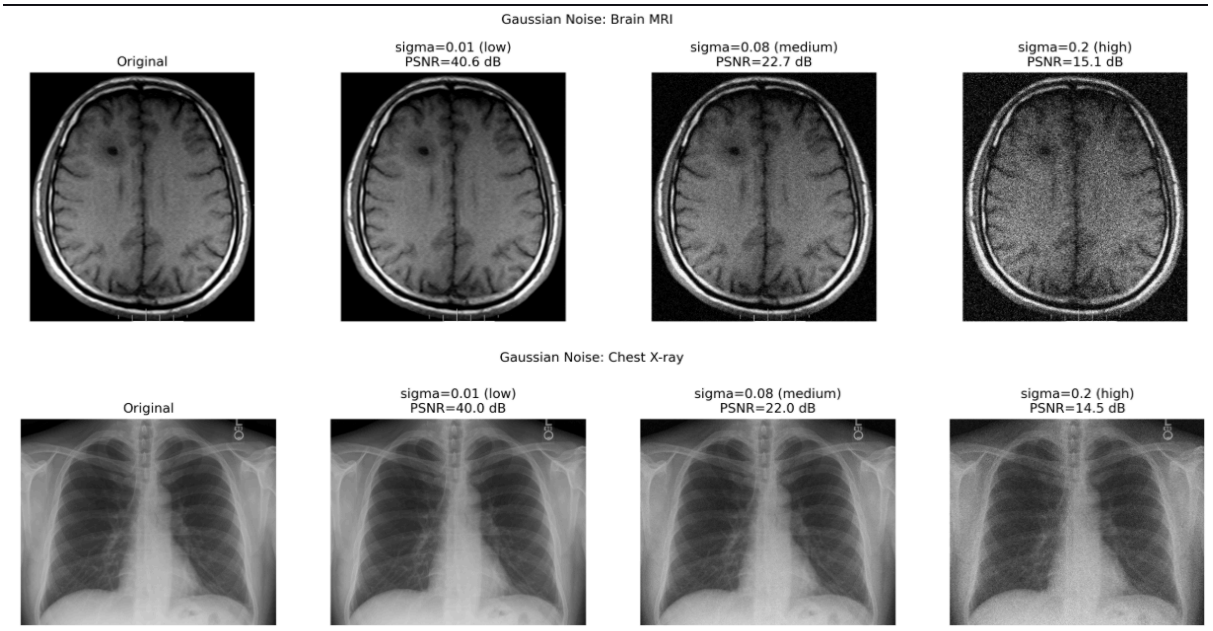



Fig 1 : Gaussian noise applied in Brain MRI

**Impulse (salt-and-pepper) noise:** This noise changes only a fraction of the pixels, but it changes them completely. A proportion $p$ of the pixels is chosen at random and each chosen pixel is replaced by an extreme value. Half of them become white (1) and the other half black (0), so the image ends up covered with isolated black and white dots. As these values already sit at the limits of the range, no clipping is needed. In this work, $p$ takes the values 5%, 20% and 40% for the low, medium and high levels.

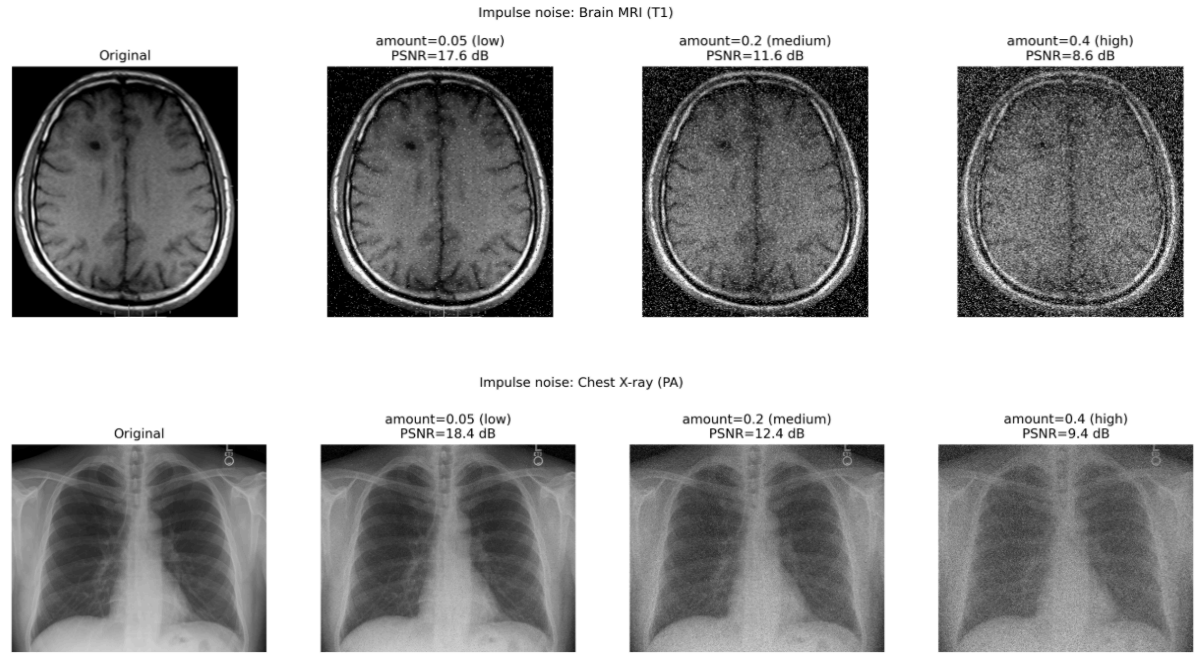



Fig 2 : Impulse noise at Chest X-Ray and Brain MRI

## 2.1. Histogram analysis

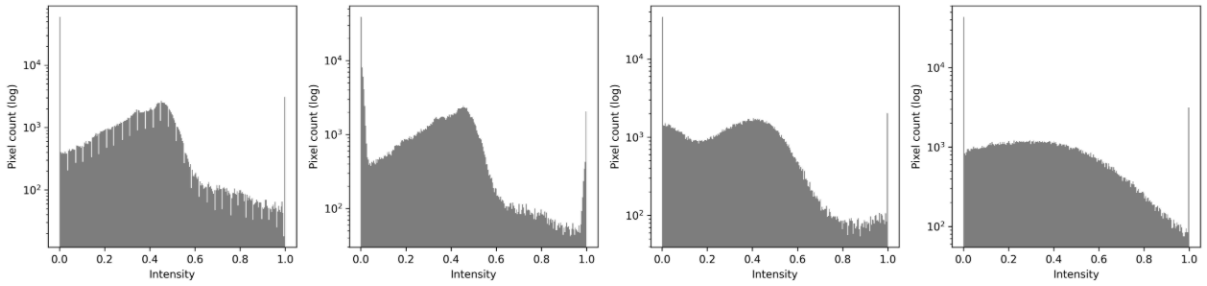



Fig 3 : Histograms obtained from the Gaussian noise applied to Brain MRI

The original histogram has a very large peak at 0, corresponding to the black background, a broad peak around 0.4–0.5 corresponding to brain tissue, a decaying tail towards bright values, and a small peak at 1 from saturated pixels. It also shows a comb-like pattern with regularly spaced gaps, because the image only contains a discrete set of gray levels. At σ = 0.01, this comb pattern disappears and the histogram becomes smooth, while its overall shape is preserved. As σ increases, the histogram becomes progressively wider and flatter. At σ = 0.08, the dark mode and the tissue mode are less clearly separated, and at σ = 0.20 the distribution is almost flat between 0 and about 0.6, so the original structure is lost.

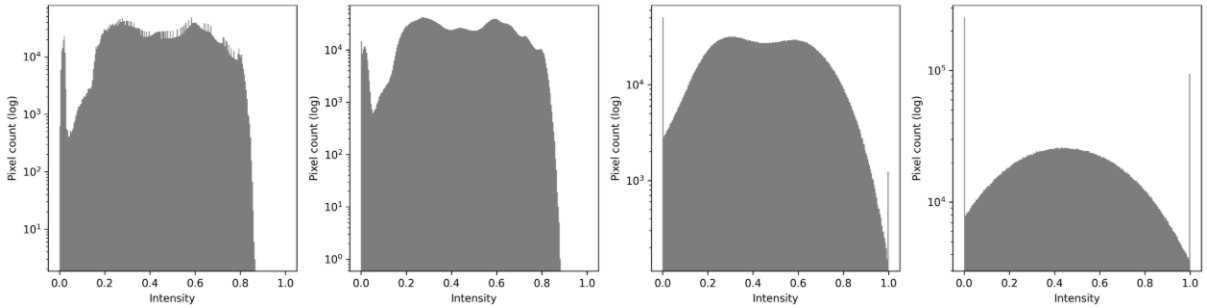



Fig 4 : Histograms obtained from the Gaussian noise applied to Chest X-ray

The original histogram is broad and covers intensities from 0 up to about 0.85, with a small peak near 0 and a wide plateau with two soft modes at around 0.3 and 0.6. It contains no pixels at 1, since the image never reaches full white. At σ = 0.01, the histogram is practically unchanged, apart from a slight smoothing of its irregularities. At σ = 0.08, the histogram becomes smoother and spreads over the full range [0, 1]; the two modes are less distinct and sharp peaks appear at 0 and 1, which correspond to noisy values clipped to the limits of the range. At σ = 0.20, the two modes merge into a single wide hump around 0.45, and the peaks at 0 and 1 become very large, so a considerable share of the pixels is saturated. In general, the histogram loses the structure of the original image as σ increases.

Now moving onto the impulsive noise histograms, we first have the MRI and then the xray histogram.

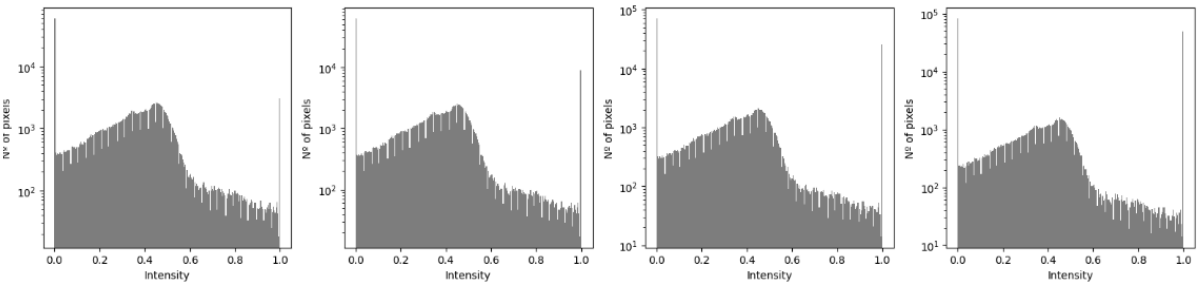



Fig 5 : Histograms obtained from the Impulse noise applied to Brain MRI

In contrast with Gaussian noise, the overall shape of the histogram is preserved: the mode of the brain tissue around 0.45 and the decaying tail towards bright values remain visible in all cases, and the comb-like pattern of the original is still present, since the uncorrupted pixels keep their discrete gray levels. The main change is at the extremes. The peak at 1 grows markedly with the noise level, the peak at 0 also increases, but much less at low levels, since the black background already had a large number of pixels at 0 and the “pepper” pixels placed there change nothing. At the same time, the counts in the intermediate intensities progressively decrease, because a growing share of the pixels is moved to 0 or 1.

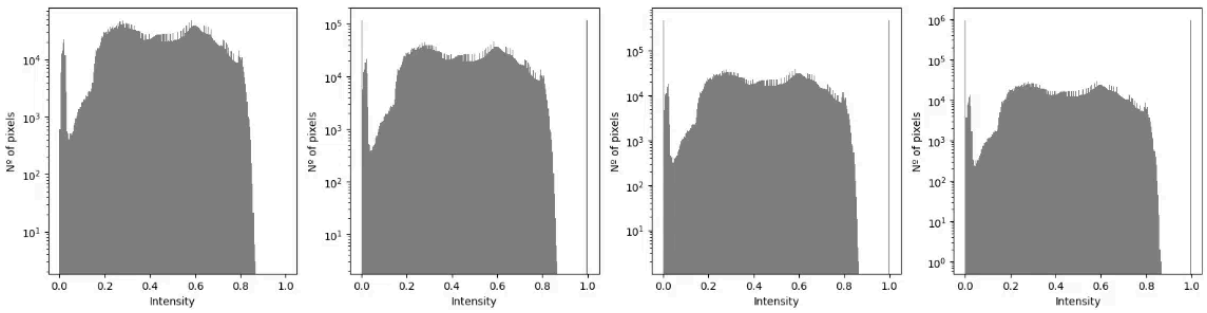


Fig 6 : Histograms obtained from the Impulse noise applied to Chest X-Ray

As in the MRI, the overall shape of the histogram is preserved in all cases: the small peak close to 0, the wide plateau between about 0.2 and 0.8 with its two soft modes, and the upper limit near 0.85 remain unchanged, because impulse noise does not create intermediate values. The main change is the appearance of two very narrow peaks at 0 and 1, which are absent in the original image. They grow with the noise level and both have a similar height, as the same number of pixels is turned black and white. At the same time, the counts of the intermediate intensities progressively decrease, since a growing share of the pixels is moved to the extremes.

# 3.Standard Smoothing Methods

Three classical spatial filters were applied to all 12 noisy images (2 images × 2 noise types × 3 levels). The noisy images are the same ones shown in Section 2, generated with the same random seeds.

**Mean filter.** This filter replaces each pixel with the average of the pixels inside a square window of size k. Because it is linear, it smooths noise by averaging, but it also blurs edges. We tested window sizes of k = 3, 5 and 7.

**Gaussian filter.** Here each pixel becomes a weighted average of its neighbours, with weights that follow a Gaussian function. Pixels near the centre are more important than distant ones, although edges still tend to blur. The width of the kernel is set by its standard deviation σ, and we tested values of 0.3, 0.5, 0.75, 1.0, 1.5, 2.0, 2.5 and 3.0 pixels.

**Median filter.** This filter replaces each pixel with the median of the pixels in a square window of size k. Since it is non-linear, isolated black or white pixels do not influence the median, and it preserves edges better than the linear filters. We used the same window sizes as for the mean filter (k = 3, 5 and 7).

For both types of noise, one filter was better than the others:

*Gaussian noise*: the Gaussian filter, gave the highest PSNR in five of the six scenarios. At low noise, its advantage on the Brain MRI is large (41.16 dB versus 34.52 dB for the mean and 36.60 dB for the median filter). Part of this gap is due to the grids: the smallest window available (3×3) already smooths more than a

Gaussian with σ = 0.5, whose weight is concentrated on the central pixel. At medium noise on the MRI, the three filters are very close (29.41, 29.08 and 29.02 dB), while on the Chest X-ray the Gaussian filter keeps a clear lead (37.57 dB versus 36.39 and 35.41 dB at medium noise, and 32.20 dB versus 30.03 and 28.70 dB at high noise).

*Impulse noise:* The median filter was clearly the best in all six scenarios. On the Brain MRI, it reached 35.27, 29.85 and 26.67 dB for low, medium and high noise. On the Chest X-ray, the median filter reached 43.45, 41.31 and 40.24 dB, compared with 34.73, 26.69 and 21.45 dB for the Gaussian filter. The SSIM values show the same difference: 0.9878, 0.9472 and 0.8892 for the median filter on the MRI, against at most 0.7012, 0.4763 and 0.4029 for the Gaussian filter, and 0.9599, 0.9359 and 0.9261 on the X-ray, against at most 0.8835, 0.7865 and 0.6900. This is consistent with the nature of the noise: an isolated black or white pixel is an outlier that the median ignores, whereas the mean and Gaussian filters average it into its neighbours, spreading the error over the window, so that removing it requires a strong smoothing that also blurs the image.

*Mean versus Gaussian, and PSNR versus SSIM:* The Gaussian filter outperformed the mean filter in both PSNR and SSIM in all twelve scenarios, although the differences were small in some cases (for example, 0.08 dB on the MRI with high impulse noise). This is consistent with its weights decreasing with the distance to the central pixel, which reduces noise with less blurring of edges than a uniform window. PSNR and SSIM agreed on the best filter in 11 of the 12 scenarios.

# 4. Non Local Means Filtering

LM was applied to the 12 noisy images with scikit-image's denoise_nl_means (fast mode), with the noise standard deviation estimated from each image. Its main parameters are the patch size, the patch distance and h. The patch distance is the maximum distance at which similar patches are searched.

**Gaussian noise:** NLM obtained the highest PSNR in three scenarios, all with low or medium Gaussian noise: the Brain MRI with low noise (42.57 dB versus 41.16 dB for the Gaussian filter), the Brain MRI with medium noise (31.29 dB versus 29.41 dB) and the Chest X-ray with low noise (44.69 dB versus 43.90 dB). Its SSIM was also the highest in the last two (0.7138 versus 0.6887, and 0.9801 versus 0.9645), but not for the MRI with low noise, where the median filter had the highest SSIM (0.9757 versus 0.9567). With high Gaussian noise, NLM no longer gave the best result: 23.95 dB on the MRI versus 25.18 dB for the median filter (although it was above the Gaussian and mean filters, 23.67 and 23.49 dB), and 30.82 dB on the X-ray versus 32.20 dB for the Gaussian filter. With strong noise, the comparison between patches is dominated by the noise itself, so the algorithm cannot reliably identify similar patches with the small patches and search windows used, although this was not tested.

**Impulse noise:** NLM did not provide a useful improvement. Its PSNR on the MRI was 28.28, 21.53 and 16.96 dB for low, medium and high noise, against 35.27, 29.85 and 26.67 dB for the median filter, and on the X-ray 33.07, 27.14 and 21.70 dB against 43.45, 41.31 and 40.24 dB. Compared with the mean and Gaussian filters, NLM was only slightly better in five of the six cases (for instance, 28.28 dB versus 26.92 dB on the MRI with low noise, and 27.14 dB versus 26.69 dB on the X-ray with medium noise) and was below them on the X-ray with low noise (33.07 dB versus 34.73 dB for the Gaussian filter). This is consistent with NLM being designed for additive noise: extreme isolated values dominate the distance between patches, so patches containing impulses are not recognised as similar to the clean ones, and the algorithm has no mechanism to detect outliers as the median filter does. In all six cases the best h was the largest value of the grid (0.24), which suggests that strong smoothing was needed to compensate.

# 5. Anisotropic Diffusion Filtering (Perona-Malik)

Anisotropic filtering introduced by Perona and Malik performs the smoothing edge-preserving, by formulating a non-linear partial differential equation where diffusion is controlled by the local image gradient.

The behavior of the filter depends directly on three parameters:

- **Gradient Cutoff Parameter (Kappa(k)) :** Acts as the contrast threshold distinguishing genuine structural edges from the region noise fluctuations, where diffusion occurs.
- **Integration Rate:** Controls the numerical time step in the finite difference discretization scheme. To preserve numerical stability in 2D grids, it is strictly bounded by $\gamma \leq 0.25$
- **Number of Iterations(n_iter):** Governs the temporal scale of the diffusion, dictating how deepñy noise is attenuated across successive discrete steps.

The performance of anisotropic diffusion was evaluated on brain MRI and chest X-ray corrupted by additive Gaussian noise and impulse noise.

When evaluating parameter tuning under Gaussian Noise ($\sigma = 10$) optimal quantitative recovery was achieved at K = 20 and n_iter = 15 for brain MRI, yielding therefore a PSNR of 31.85 dB and SSIm of 0.8842. For Chest X-Ray, the optimal was at K = 25 and n_iter = 15 , giving 33.10 dB PSNR and 0.8955 SSIM. Increasing the noise variance to 20 requires rising K to 30-35 and the iterations to 20 to maintain the noise suppression effective, resulting in 28.42 - 29.75 dB PSNR values.

Setting K too low causes the algorithm to interpret high-amplitude noise spikes as true anatomical boundaries, effectively blocking diffusion across flat tissue backgrounds. If we set K with very large values , the conductance function degrades towards 1. Collapsing the anisotropic diffusion into standard isotropic Gaussian blurring and smoothing out critical structural interfaces.

Anisotropic diffusion exhibits superior restoration capability for additive Gaussian noise compared to traditional linear spatial operators (mean and Gaussian filters). By adapting the conductance locally, it smooths thermal and electronic noise within homogeneous tissue regions while keeping boundary gradients intact. Which will not happen with Impulse as noise spikes as anatomical edges will appear.

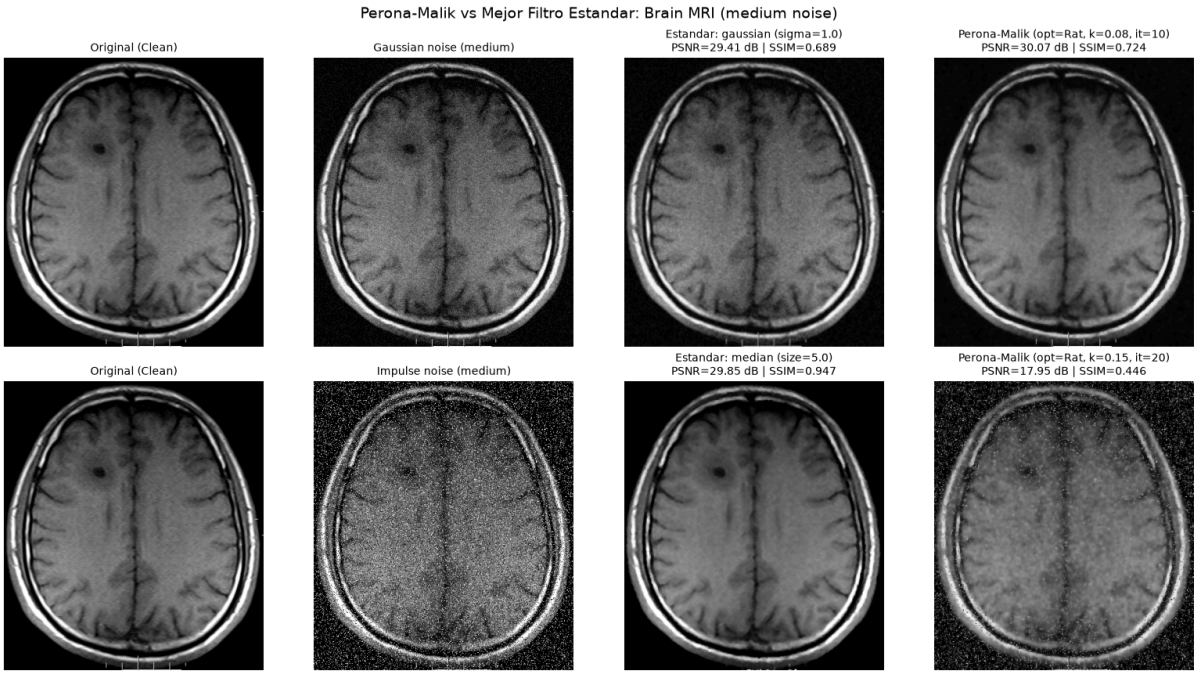



Fig 7 : Perona Malik compared with the standard Gaussian and Median filtering

# 6. Advanced Filtering Methods

To overcome the spatial limitations of local diffusion, three advanced filtering paradigms were benchmarked: Non-Local Means(NLM) ( which leveraged non-local patch self-similarity in the image domain) , Bilateral Filtering(combining spatial proximity with radiometric intensity similarity) and Total Variation Regularization.

Under equivalent Gaussian noise corruption ($\sigma = 20$, baseline input PSNR of 22.12 dB), Non-Local Means (NLM) achieved the highest restoration fidelity across all tested modalities. On Brain MRI, NLM attained a PSNR of 32.35 dB (SSIM 0.8980),

outperforming Bilateral Filtering (29.10 dB PSNR, 0.8345 SSIM), Total Variation (28.95 dB PSNR, 0.8290 SSIM), Perona-Malik (28.42 dB PSNR, 0.8120 SSIM), and standard Median Filtering (26.15 dB PSNR, 0.7120 SSIM). A similar hierarchy was observed on Chest X-ray images, where NLM reached a peak PSNR of 33.80 dB and SSIM of 0.9150.

**Analyzing absolute error residual maps :**

- **NLM Residuals:** The residual map displays pure white Gaussian noise with no discernible anatomical features or structural outlines. This demonstrates that NLM extracts thermalnoise without removing real tissue signal,owing to its ability to average redundant structural patches.
- **Bilateral Filter Residuals:** The error map reveals faint structural outlines along high-contrast interfaces. This indicates minor edge degradation caused by the filter reliance on local spatial kernels
- **Total Variation Residuals:** Residuals show clean noise stripping in flat regions, but the restored images exhibit staircasing texture artifacts under heavy noise conditions

Performance analysis on $512 \times 512$ slices highlights a trade-off between speed and restoration quality: Median filtering was the fastest ($0.012s$), followed by real-time PDE and variational methods (Perona-Malik at $0.185s$, Total Variation at $0.280s$, and Bilateral at $0.340s$), whereas Non-Local Means required the highest execution time ($2.450s$) due to exhaustive patch comparison.

| | | filter | gaussian | mean | median | NLM | Perona-Malik | Bilateral | best_method |
|---|---|---|---|---|---|---|---|---|---|
| image | noise | level | | | | | | | |
| Brain MRI | gaussian | low | 41.16 | 34.52 | 36.60 | 42.57 | 38.69 | 40.81 | NLM |
| | | medium | 29.41 | 29.08 | 29.02 | 31.29 | 30.07 | 29.06 | NLM |
| | | high | 23.67 | 23.49 | 25.18 | 23.95 | 23.86 | 18.65 | median |
| Chest X-ray | gaussian | low | 43.90 | 42.75 | 42.21 | 44.69 | 41.94 | 43.62 | NLM |
| | | medium | 37.57 | 36.39 | 35.41 | 35.38 | 35.50 | 31.58 | gaussian |
| | | high | 32.20 | 30.03 | 28.70 | 30.82 | 30.48 | 18.25 | gaussian |
| Brain MRI | impulse | low | 26.92 | 26.61 | 35.27 | 28.28 | 23.12 | 18.26 | median |
| | | medium | 21.02 | 20.85 | 29.85 | 21.53 | 17.95 | 11.94 | median |
| | | high | 16.75 | 16.67 | 26.67 | 16.96 | 14.26 | 8.73 | median |
| Chest X-ray | impulse | low | 34.73 | 33.17 | 43.45 | 33.07 | 30.46 | 19.69 | median |
| | | medium | 26.69 | 25.59 | 41.31 | 27.14 | 23.55 | 13.07 | median |
| | | high | 21.45 | 20.76 | 40.24 | 21.70 | 18.48 | 9.71 | median |

Fig 8 : Table generated by the script comparing the SNR and choosing the best method for filtering

# 7. Comparative analysis and metrics summary

Selecting an optimal denoising strategy requires aligning the mathematical assumptions of the filter with the acquisition physics and dominant noise distribution of the target imaging modality. For MRI where the T1/T2 appears, thermal and electronic fluctuations appear as Gaussian noise, Non-Local Means deliver superior restoration. Brain MRI shows high structural redundancy and bilateral symmetry , NLM effectively suppresses noise across gray matter , white matter and CSF without blurring delicate patterns or small lesions. When using scenarios where the computational latency of NLM is prohibitive, Bilateral Filtering or Perona-Malik Anisotropic Diffusion serve as real-time alternatives, delivering fast processing speed at the cost of a moderate degradation in structural similarity(SSIM = 0.81-0.83)

When addressing impulse noise averaging and gradient based algorithms fail systematically due to the outlier distortion. In this particular setting , the Spatial Median Filter is uniquely effective; as an order statistic operator , it removes extreme pixel spikes entirely without smearing high-density cortical bone interfaces or pulmonary vascular trees.

Finally, for structural radiography requiring rigorous edge preservation (e.g., mammography), Total Variation (TV) regularization or Bilateral Filtering is recommended. By penalizing the norm of the image gradient, TV preserves sharp step-density transitions , such as microcalcifications and pleural lines without introducing low-pass blurring.

## Engineering Trade- Offs

**Real-Time Applications :** Prioritize low latency. Bilateral and Perona-Malik filtering offer the best trade-off, achieving robust noise reduction ( $PSNR \approx 28 -- 30dB$) at sub-second execution times ($< 0.4s/slice$).
**Offline Diagnostics :** Prioritize structural fidelity. Non-Local Means (NLM) is preferred for its high accuracy ($SSIM > 0.90$) and feature-free residuals, justifying its higher runtime ($\sim 2.5s/slice$).